

# Two Coastal-Margin Blockers in the AWI-ESM3 ↔ PISM Moving-Cavity Coupling

A measurement-driven investigation log

- I.** OIFS step-0 coastline crash (*resolved*)    •    **II.** FESOM cavity-margin vertical-CFL instability  
(*root cause: a geostrophically-unbalanced remapped state — restart-value fixes tested & excluded*  
— §1, read first)

Experiment `movcav4`, chunk-3 (`awiesm3` 1901)

2026-07-05

## Abstract

We are building an end-to-end AWI-ESM3 (FESOM2 + OIFS-48r1 + LPJ-GUESS) ↔ Antarctic-PISM *moving-cavity* coupled run driven by `esm-tools`. On each mesh-change leg the coupling machinery regenerates the FESOM sub-mesh, the OASIS grids/masks/weights, and the OIFS ICMGG initial file so all components agree on the new coastline. The first mesh-change leg (`movcav4`, `awiesm3` 1901) surfaced **two distinct blockers, both localised to the Antarctic coastal margin**. This log records the investigation as a sequence of *measurements* rather than hypotheses.

**Blocker I — an OIFS step-0 surface crash — is SOLVED.** A floating-point NaN in the surface/moist-physics column struck exactly the coastline cells whose land-sea type flipped. Root cause: the coupling tool `ocp-tool` flipped a cell by setting *only* its land-sea mask and soil type, leaving every other surface field at the old type (a “half-converted” column); OIFS faithfully re-ingested that inconsistency from the regenerated ICMGG and NaN’d in the moist physics. Fix: a masked nearest-neighbour rebuild of each flipped cell in `ocp-tool`, plus the ICMGG re-ingest on warm start. A residual crash then proved *DDH-diagnostic-only*; with DDH off, OIFS runs ~700 coupling steps under `-fpe0`, so OIFS and the fluxes it sends the ocean are finite. Part II (§5 ff.) is the full record.

**Blocker II — a FESOM vertical-CFL blowup — is the LIVE blocker, now root-caused.** Some 5–10 model days in, FESOM blows up with a vertical-CFL runaway ringing the cavity margin. The cause is a **geostrophically-unbalanced remapped state**: the mesh change plants fresh cavity fronts against salty open water but *zeroes* the balancing current, so the model’s  $t = 0$  adjustment grows  $w$  at the fronts. This is the *same* instability the hourly movie showed *growing over days* — a  $t = 0$  imbalance that amplifies, not a melt process from a balanced start. Every restart-*value* fix (freezing cap; coherent  $T/S$  refill to zero density inversions) was tested and does **not** prevent it; a 60s step clears the *fast* mode but a *slow* barotropic `eta_n` mode then grows. Fix: initialise the changed-cell velocity in balance. **§1 is the current truth — read it first.**

## 1 Current state (2026-07-06, rev. 3): the blowup is a *dynamical imbalance* of the remapped state, not a restart-value pathology

**Read this first — supersedes §2 (rev. 2).** The rev. 2 basal-melt / thin-layer story led to two concrete restart fixes — a freezing-point cap, then a full coherent-column  $T/S$  refill. Both are now **implemented and tested** on a clean leg (`movcav8`, `awiesm3` 1901); together they make the remapped restart **provably clean** (zero static density inversions anywhere) — yet **neither prevents the blowup**. The true root is that the remap hands FESOM a state that is **out of geostrophic balance at the moving margins**: real (and fill-sharpened) lateral density fronts carried across the mesh change, with the balancing current **zeroed out**. That single imbalance expresses *fast* as the vertical-CFL runaway (1200s) and *slow* as a barotropic `eta_n` runaway (60s).

## 1.1 What was tested since rev. 2, and what it falsified

1. **Freezing-point cap — applied, no effect.** `cap_new_cavity_temp` ran in `movcav8`'s `couple.in` (1968 nodes / 14 670 cells capped to  $T_f$ ). The leg still blew (`mstep`≈205, `CFLz`≈5.7). Removing the thermal driving does *not* stop it.
2. **Coherent-column  $T/S$  refill — all inversions removed, still blows.** The per-level unmasked nearest-node fill (which planted melt-fresh  $S \approx 22$  mid-column) was replaced by (a) a *mask-aware open-ocean donor* for surface-opened columns and (b) an *own shelf-base hold* for still-sub-ice retreated columns (committed, `ad4e84b2`). Over all 1595 retreated cells the worst static stability improved  $-8.8 \rightarrow -0.5 \text{ kg m}^{-3}$  (11 severe  $\rightarrow 0$ ). The leg *still* blew (`mstep`≈164, `CFLz`≈8.5), essentially unchanged. **Restart  $T/S/\text{ssh}/\text{density}$  is decisively excluded** (four inversion-free variants all blew the same way).
3. **Timestep = 60 s — clears the fast blowup, exposes a slow one.** With `step_per_day`=1440 (everything else identical) the run made 0 `CFLz` warnings and reached `mstep`= 14 629 (≈10 model-days,  $4\times$  longer), then died on `eta_n`→**NaN** (barotropic `ssh`) at 112.8/−65.9. So the *fast* blowup is genuinely vertical-CFL (timestep-curable) but a *slow barotropic* mode remains — the  $\Delta t$  ramp alone is **not** a cure.

## 1.2 The initiation mechanism (fast blowup): unbalanced fronts + zero velocity

The updated plots (`movcav8` v4 restart; PolyCollection on the real mesh triangles, dateline-wrapping triangles filtered, layer-index vertical axis so the thin cavity layers are visible; `movcav4_crash_plots/initstate/`, Figs. 1, 2) show it directly. Around the *earliest* `CFLz` seed (Weddell, −62.96/−66.81, first warning `mstep` 4, peak level 11  $\approx 95 \text{ m}$ ):

- **A picket-fence of fresh cavity columns.** Opened cells (ice retreated) come out **uniformly fresh**,  $S \approx 31$ , punched into an  $S \approx 33\text{--}34$  background — strong **lateral density fronts at the grid scale ( $\sim 25 \text{ km}$ )**, strongest mid-column. Advanced cells (open→new ice) keep salty  $\sim 34$  water and are comparatively benign — so the *opened* cells are the troublemakers.
- **Velocity is exactly zero across the whole opened region** — the remap zeroes  $u/v$  on every changed element; the speed panels are solid black except the far-field unchanged edge.
- A density front *with zero velocity* is **not in thermal-wind balance**: at  $t = 0$  the model must spin up the missing geostrophic current, and its **ageostrophic part is  $w$** . The  $w$  panels show exactly that — a down/upwelling dipole **pinned to the fronts**, mid-deep layers. At  $\Delta t = 1200 \text{ s}$  over  $\sim 10 \text{ m}$  layers this  $w$  ( $\sim 2 \text{ mm s}^{-1}$ ) gives `CFLz`> 1 **within  $\sim 4\text{--}10$  steps**.

This is precisely why the  $T/S$  *stability* fixes did nothing: the imbalance is **lateral (fronts) + missing momentum**, not a vertical inversion. The *over-freshening* is partly self-inflicted — the shelf-base “hold upward” plants the freshest (ice-base) water throughout the newly-opened column, sharpening the very fronts that drive the adjustment.

## 1.3 Peak `CFLz` is a consequence, not the seed

The *highest-CFLz* cell at the crash (90.5/−63.5, level 38, **3775 m**, `CFLz` 8.45) is **deep, fully-unchanged open ocean with no cavity at all**, and a perfectly smooth initial state ( $S$  varies 0.02 PSU; Fig. 2, top). It is a *late* manifestation: the instability seeds at the opened margins and **advects/propagates into the deep basin by the crash step**. So “high `CFLz` at the crash”  $\neq$  “where it started” — only the earliest (margin) cells show the initiating imbalance.

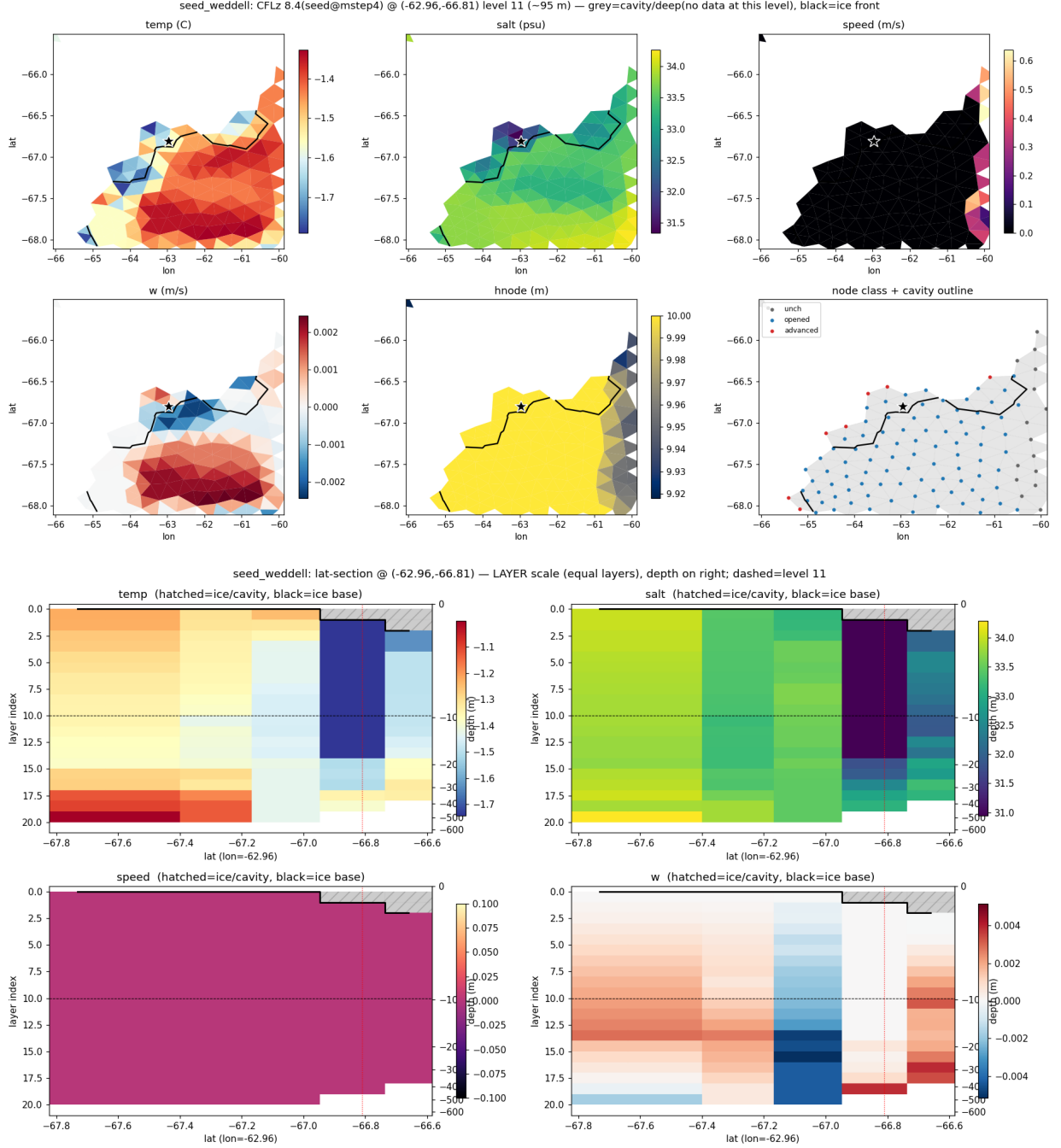


Figure 1: **Initiation seed — Weddell margin** ( $-62.96/-66.81$ ), **movcav8 v4 restart**. *Top*: plan view at  $\sim 95$  m (level 11). Note the isolated **fresh column** ( $S \approx 31$ ) at the ice front (black outline), the **all-black speed panel** ( $u/v = 0$  across the opened region), and the  $w$  dipole on the front. Grey = cavity/deep (no data at this level, *not* land). *Bottom*: lat-depth section on a **layer-index axis** (equal layers; depth on the right) — the fresh column runs full-depth under a thin hatched ice cap, against  $S \approx 34$  open water, with the  $w$  down/upwelling curtain at the front. The state has strong fronts but *zero* balancing flow.

## 1.4 The slow (60 s) blowup is the same root, a different mode

At 60 s the vertical CFL stays subcritical, so instead a **barotropic  $\eta_n$  response accumulates over  $\sim 10$  days** and piles against the coast/slope *adjacent* to the changed margins. The 60 s crash site (112.8/−65.9, Fig. 2, bottom) is a well-stratified continental slope (warm CDW at depth, surface jet) — a *benign* local state, but only 32 km from the advanced-ice margin (9 changed cells within 100 km). So  $\eta_n \rightarrow \text{NaN}$  there is a **slow remote pile-up**, not a local shock. Fast (CFLz) and slow ( $\eta_n$ ) are **two faces of one imbalance**.

## 1.5 Reconciling the two views: a $t = 0$ imbalance that *amplifies*

The rev. 2 hourly movie (movcav4; Figs. 3, 4) and this step-1 analysis (movcav8) are the *same* instability at two times; they reconcile once “clean” is read correctly:

- **“Hour 1 clean” meant clean *values*, not a balanced state.**  $T/S/\text{ssh}$  are in normal ranges with no NaNs, but the state is out of geostrophic balance (fresh margin fronts vs salty open water,  $u/v=0$ ). The movie’s “ $w \approx 0$  at hour 1” is that zero initial velocity.
- **The day-scale “growth” is the adjustment of the  $t = 0$  imbalance**, not a new process — the missing geostrophic current spins up,  $w$  amplifies at the fronts, the fresh/cold band saturates and spreads: exactly the movie’s onset→runaway.
- **Melt is a secondary amplifier, not the driver.** rev. 2 pinned the freshening on melt under new ice (*advanced* cells); the fresh fronts are largely a remap-fill feature at the *opened* cells, present at  $t = 0$ , and the freezing cap (which removes the melt driving) does not stop the blowup.

So rev. 2 read the *character* right (a fresh-fronted  $\sim 95$  m coastal baroclinic mode that grows over days) but mis-read normal restart *values* as a balanced IC and over-credited melt. The unifying cause is the unbalanced remapped state.

## 1.6 Root cause and ranked fix

On the spun-up full mesh these fronts carry their balancing geostrophic currents; the remap preserves  $T/S$  (density) but **discards the momentum field on changed cells (zeroes  $u/v$ ) and over-sharpens the opened columns**, handing the model a state far from balance. It must then adjust — fast (vertical CFL) or slow (barotropic). Ranked:

1. **Initialise  $u/v$  in thermal-wind balance with the remapped density** (or copy the neighbouring unchanged current onto changed elements) instead of zeroing. Removes the root; addresses *both* modes. **Highest leverage.**
2. **Do not over-fresh the opened columns** — blend toward the adjacent open-ocean profile to soften the grid-scale fronts.
3.  **$\Delta t$  ramp** for the mesh-change leg — necessary but *not sufficient* (60 s cleared the fast mode but the slow barotropic one still killed it at day 10); must be combined with (1)/(2).

Everything observed — fronts + black speed panels +  $w$  dipoles on the fronts, fast and slow — points to (1): give the changed cells a *balanced* velocity instead of zero.

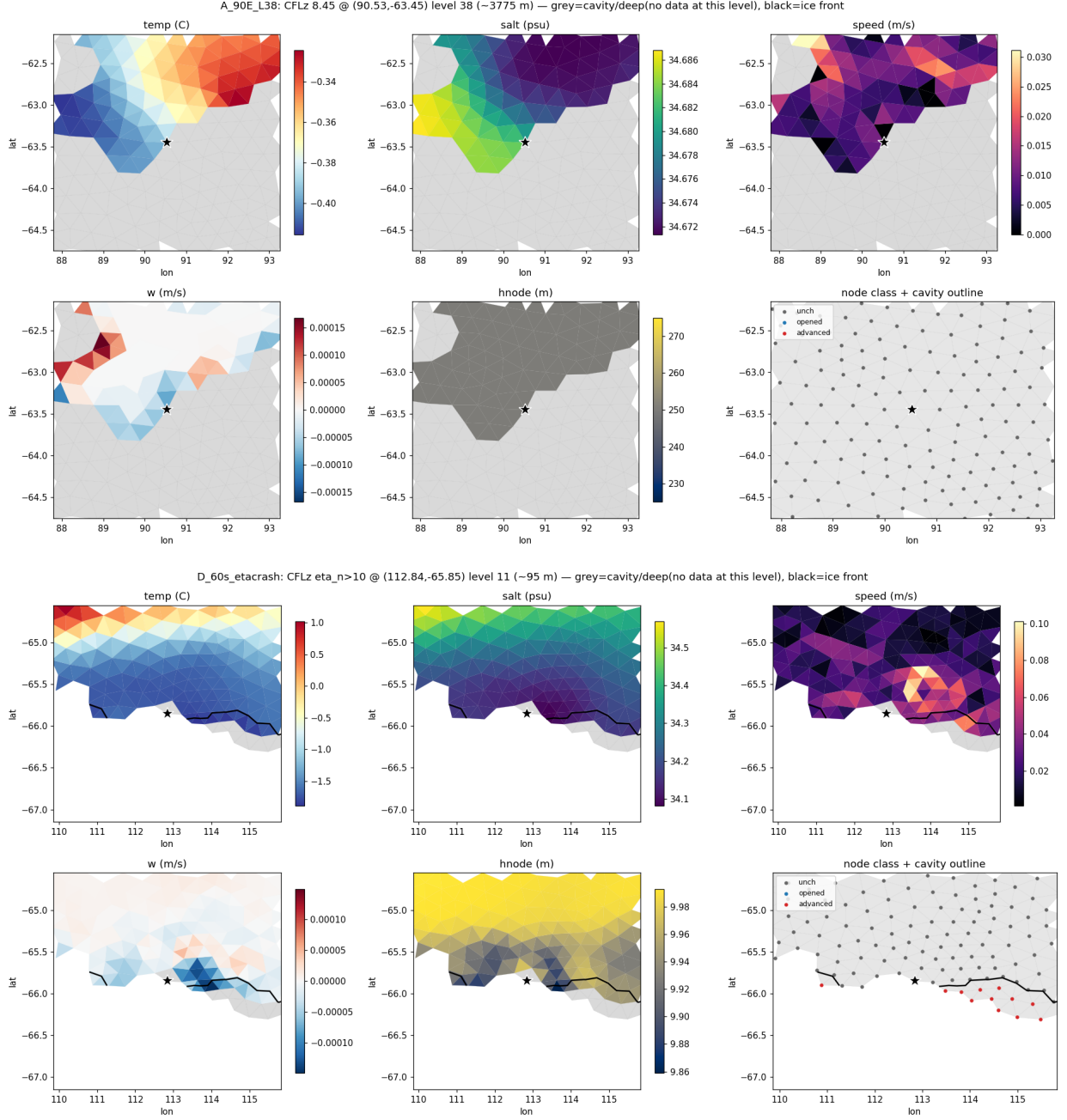


Figure 2: **Two other high-CFLz sites — both benign at  $t = 0$ , confirming the seed is elsewhere.** *Top:* the peak-CFLz cell (90.5/−63.5, level 38, **3775 m**) is **fully-unchanged deep open ocean** ( $S$  varies 0.02 PSU) — a late consequence of the instability spreading, not a cause. *Bottom:* the **60s eta\_n crash** site (112.8/−65.9) is unchanged coastal slope 32km from the advanced-ice margin — a mild state where the barotropic mode piles up over  $\sim 10$  days.

## 2 Current state (2026-07-05, rev. 2, *superseded by §1*): a *time-growing* coastal baroclinic instability on the submesh — *narrowed, not yet root-caused*

**Read this first.** The OIFS step-0 surface crash that the rest of this document dissects (§8.5 onward) turned out to be a *self-inflicted regression* (a stale, pre-NN-fill ICMGG); with the correctly regenerated ICMGG the OIFS side **works**. The live blocker is that the coupled chunk-3 leg (movcav4, awiesm3 1901) runs ~5–10 model days and then **FESOM blows up** with a vertical-CFL runaway. *This section is the current truth and supersedes §8.5ff.* Today’s work **narrowed the cause substantially**: it is **not** the timestep, **not** the surface flux/ice, and **not** a corrupt or fresh-capped remapped restart. Hourly output plus a targeted diagnostic run point strongly to a **basal-melt shock where the moving geometry puts new ice over above-freezing open-ocean water** (§2.4). A candidate fix is drafted (freezing-point cap on newly-under-ice columns) but has **not yet been tested** in a run.

### 2.1 Symptom

A vertical-CFL ( $\text{CFLz} = w \Delta t / \Delta z$ ) **instability that rings the entire Antarctic ice-shelf margin and grows over days from a clean start**. First  $\text{CFLz} > 1.75$  warnings appear as early as  $\text{mstep} \approx 11$  at ~29 fixed coastal/cavity locations;  $\text{CFLz}$  creeps  $1.8 \rightarrow 2 \rightarrow \dots \rightarrow$  tens over days; eventually a cell’s temperature goes NaN (“found temperture becomes NaN”) and FESOM MPI\_ABORTs. The blowup *step* is stochastic (MPI/threading): at **1200 s** it blew at  $\text{mstep}$  311–696 (4.3–9.7 d; runs 26060316 seeded / 26055476 unseeded), and the hourly-output run 26061910 at  $\text{mstep} = 608$ . The *hotspot pattern* is robust and *subsurface* —  $w$  and  $\text{CFLz}$  peak **mid-column at ~95–100 m**, not at the surface. Hotspots (glon/glat, peak  $\text{CFLz}$ ): Totten **108.7/–66.1** (~8), Amery/Prydz 71.4/–70.0, Ross 165.6/–75.0, Weddell/Filchner –44.3/–77.3 & –57.6/–75.9, Bellingshausen –76.5/–72.1, E-Antarctica 93.2/–66.0.

### 2.2 What we ruled OUT (with evidence)

1. **Anomalous atmosphere heat flux — NO.** Reproduced with OASIS EXPOUT (per-coupling-step dump). The coastal OIFS→ocean heat flux peaks 600–700 W/m<sup>2</sup> but is **almost entirely solar** (A\_Qs\_all; Jan solstice at –66°; flux\_decomp.py); the non-solar part (A\_Qns\_oce) is a modest –60 to –120 W/m<sup>2</sup> (normal open-water loss). Flux *leads*, SST *lags*. Forcing is physically correct.
2. **Missing sea ice ⇒ solar over-absorption — tested, NO EFFECT.** Implemented an **ice-donor NN-fill** in the F90 remap (§2.6), seeding newly-exposed nodes with the nearest iced neighbour’s ice column (run 26060316, verified 47 455 iced). The  $\text{CFLz}$  hotspots came back **location-, onset- and magnitude-identical** to the unseeded run. Ice changed *nothing* — the instability is **ocean-internal**.
3. **Timestep / vertical-CFL artifact — revised by the 60 s test (§1): the fast blowup is vertical-CFL; 600 s was simply not small enough.** Halving to **600 s** (step\_per\_day=144, run 26061443) still blew ( $\text{mstep} = 845 \approx 5.9$  d vs the 1200 s baseline’s 4.3 d) with  $\text{CFLz}$  still 61–71 at depth — which *looked* like “not  $\Delta t$ ”. That inference was wrong: the adjustment  $w$  is still supercritical at 600 s, so the runaway regenerates. Going 20× smaller settles it — **at 60 s (step\_per\_day=1440) the leg makes zero CFLz warnings and runs ≈10 model-days** ( $\text{mstep} = 14629$ ), confirming the *fast* mode is a genuine vertical-CFL instability of the unbalanced initial state. It is **not** a full cure, though: at 60 s a *slow* barotropic **eta\_n** mode instead grows over ~10 days and NaNs (§1). So  $\Delta t$  is a necessary lever for the fast mode but must be paired with fixing the initial imbalance.

4. **Corrupt or fresh-capped remapped restart** — **NO** (and a red herring retracted). The *initial* remapped restart is **clean**: first-wet-level  $T/S$  normal ( $S$  mean 33.7, no zeros, identical to source),  $\text{hnode} \geq 0$  with sane column sums,  $\text{ssh} \in [-2.1, 1.2]$  m,  $w \approx 1.7$  mm/s at Totten, and **zero NaNs**. *Retraction*: an apparent “2649 nodes with  $\text{salt} = 0$ ” is **legitimate under-ice cavity masking** (level-index 0 at cavity nodes with  $\text{ulevels}$  2–27), *not* a fresh cap; at each node’s *first wet level*  $S$  is normal and matches the source, and *none* of those nodes retreated. The  $\sim 12$  PSU seen in the blowup-file profile is the *evolved* state (the symptom), not the initial condition. **Do not re-chase this**.
5. **Stale OASIS submesh artifacts** — **NO**. `grids.nc`, `masks.nc`, `areas.nc` (feom `srf` =  $3.608 \times 10^{14}$  m<sup>2</sup>), `rstos.nc` and the `_25` weights are all correct submesh (208 679). The only full-mesh (211 567) leftover was the namcouple feom grid-dimension line — now fixed per-leg (§2.5).

### 2.3 What the hourly movie shows (the key new evidence)

We added **hourly**, **hourly-split** output of `temp/salt/w` (§2.5) and captured the full onset→blowup. Reading a cross-margin level at  $\sim 95$  m in the Totten region (`movcav4_crash_plots/blowup.TSw_evolution.mp4`; Fig. 3):

- **Hour 1**: clean — smooth  $T/S$ ,  $w \approx 0$  everywhere.
- **~Day 2**: a **fresh + cold tongue nucleates at the retreated cavity margin** near 108.7/−66.1, with the first up/down  $w$  cells beside it.
- **~Day 3.5**: the fresh/cold band **saturates** ( $< 30$  PSU) and **spreads along the whole margin** (100–118°E); the  $w$  dipoles intensify.
- **→ runaway**:  $w$  grows until the CFLz NaN.

So the blowup is a **coastal baroclinic mode that grows during the run**, at  $\sim 95$ –100 m, **fed by progressive freshening at depth along the margin** — a *process*, not a static initial-condition pathology (hour 1 is clean; the restart is verified clean).

### 2.4 rev. 2 hypothesis: a basal-melt shock at newly-under-ice nodes (tested, superseded by §1)

The rev. 2 chain, each step measured (verify-on-output plus a diagnostic re-run with hourly `fw/fh/N2/Kv`, run 26063909): the mesh change puts new ice over above-freezing open-ocean water ( $\text{ul}_{\text{new}} > \text{ul}_{\text{old}}$  at 1878 nodes, 667 open→cavity, ringing the margin, drafts to  $\sim 490$  m); the remap copies that warm water straight under the ice; melt fires immediately (`fh`  $\sim 250 \rightarrow 2000$  W/m<sup>2</sup> band on/along the crash front; `fw`  $\sim 100$  m/yr,  $100\times$  physical); the freshwater dumped into the  $\sim 10$  m (min 4.6 m) front layer gives  $\Delta S = -4$  to  $-8$  PSU/12 h, top-down at the ice base (Fig. 4) → buoyant overturning → CFLz. **This was implemented and tested and does *not* hold as the driver**: the freezing cap `cap_new_cavity_temp` (called at the end of `remap_all_restarts`; drives the front  $T - T_f$  to exactly 0 on the 1878 nodes) *and* a full coherent  $T/S$  refill were both run in `movcav8`, and **neither prevents the blowup** (§1). Melt is a *secondary* amplifier of a state already unbalanced at  $t = 0$  (reconciliation, §1). *Structural aside, still worth a check*: deep-draft open→cavity transitions (130–490 m of ice in one 10-yr PISM step) are physically implausible — verify the submesh `cavity_nlvls` against the PISM ice field.

### 2.5 Diagnostics + infrastructure that landed (keep these)

- **Hourly + hourly-split FESOM output** of `temp/salt/w` (the enabler for the movie). `namelist.io` `io_list`: those three set to 1, 'h'; `file_def_fesom.xml`: a new `fesom_1h` file\_group with `output_freq="1h"` `split_freq="1h"`. The **hourly split\_freq is essential** — XIOS closes+flushes each hour’s file

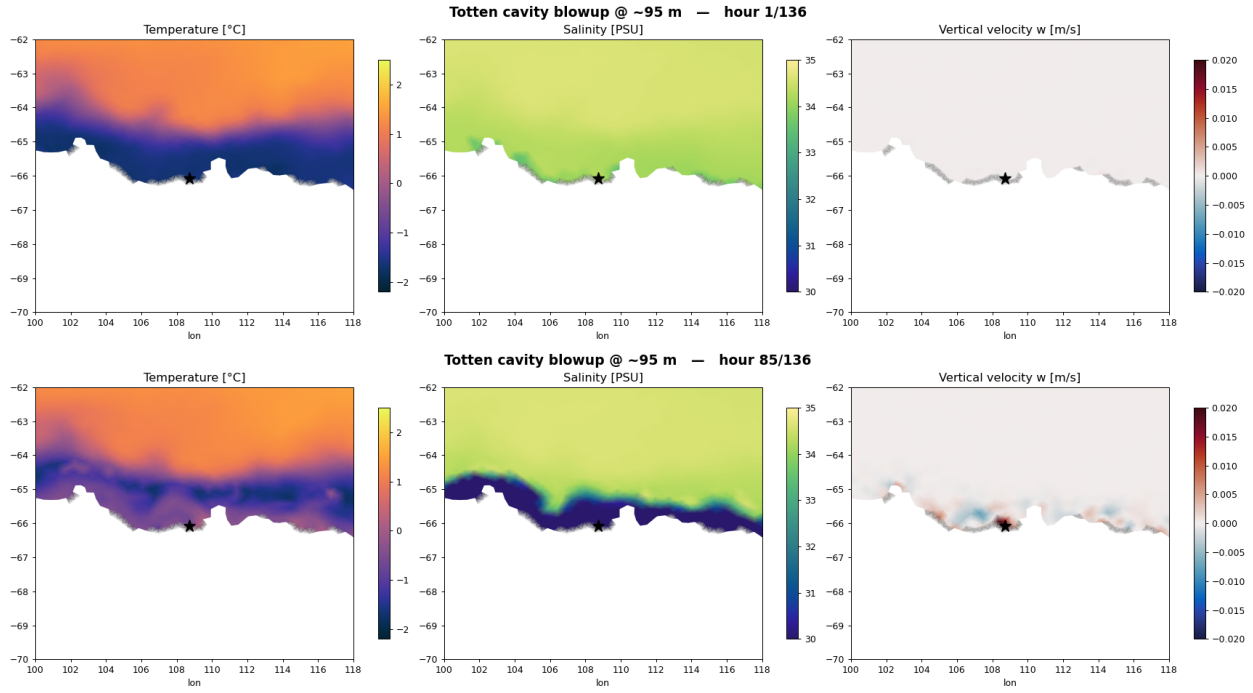


Figure 3: **Map evolution at ~95 m in the Totten margin** ( $T - S - w$ ; star = crash cell 108.7/−66.1). *Top*: hour 1 — smooth  $T/S$ ,  $w \approx 0$ . *Bottom*: hour 85 (~day 3.5) — a fresh ( $< 30$  PSU) + cold band has grown along the whole retreated margin with intensifying up/down  $w$  cells. Frames from blowup.TSw\_evolution.mp4.

to disk, so the blowup MPI.ABORT cannot lose it (a monthly/yearly split keeps one file open all run and is lost on the crash). Verified: 136 closed hourly files survived the blowup.

- **Movie tooling:** `make_movie.py` (3-panel  $T/S/w$  map at ~95 m per hour → mp4) and `make_section_movie.py` (cross-margin depth-vs-latitude section per hour, under-ice levels masked). `ffmpeg` at `/sw/spack-levante/ffmpeg-`
- **namcouple feom dimension** → submesh, per leg: `couple_namcouple subjob calling fix_namcouple_feom_dim` in `coupling_ice2fesom.interactive_mesh.functions`; `WORK_DIR.fesom` in `env_fesom.py`. Unblocks OA-SIS EXPOUT.
- **Ice-donor NN-fill** in the F90 remap (§2.6): correct, kept, but does *not* fix the blowup.
- All plots/movies → `/home/a/a270092/esm_tools/movcav4_crash_plots/`, git-ignored (\*.png, \*.mp4).

## 2.6 Reproduce / resume pointers

Experiment `movcav4`, leg `/work/ab0246/a270092/pism_repro/experiments/movcav4/run_awiesm3.19010101-19011231`. Re-run in place with `scripts/movcav4.compute.19010101-19011231.run` (it `sruns` directly in `work/`, no `prepcompute` — so edits to `work/` `namelist.*` and `*.xml` take effect as-is). Blowup runs: 26055476 (1200s unseeded, m696), 26060316 (1200s seeded, m311), 26061443 (600s, m845), 26061910 (1200s + hourly output, m608). **Hourly  $T/S/w$  files** are in `work/` (`temp.1h.fesom*`, `salt.1h`, `w.1h`) — *grab before the next leg overwrites them*. Movie/analysis scripts live in the session scratch; outputs in `movcav4_crash_plots/`. Analysis env: `/home/a/a270092/.conda/envs/pyfesom2_env/bin/python` (base python’s `matplotlib` 3.9 breaks `cmocean`; `imageio-ffmpeg` also installed there). Crash-plot suite: `/work/ab0246/a270092/software/fesom2_crash_analysis` (config points at the `movcav4` submesh + blowup file). Ice-donor code: `couplings/fesom/remap_restart/mo_remap_fields.F90` (thresholds `ICE_CONC_THRESH=0.15`, `ICE_FILL_MAXDIST=150` km; rebuild with `remap_restart/build.sh` — *delete stale \*.mod first*).



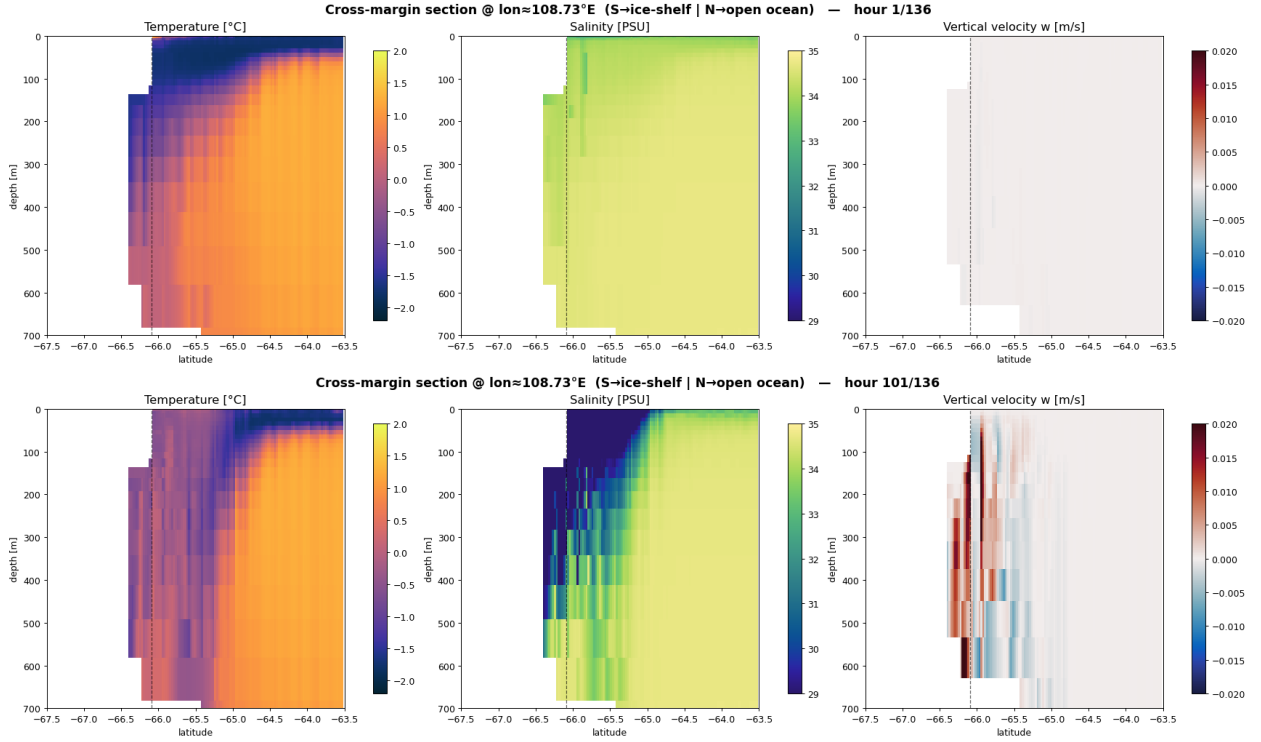


Figure 4: **Cross-margin vertical section** at  $\text{lon} \approx 108.73^\circ\text{E}$  (S  $\rightarrow$  ice-shelf, N  $\rightarrow$  open ocean; dashed line = crash latitude  $-66.09^\circ$ ). *Top*: hour 1 — cavity side is *uniform*  $\sim 34$  PSU full-depth,  $w \approx 0$ . *Bottom*: hour 101 ( $\sim$ day 4.2) — a **full-depth fresh** ( $< 29$  PSU) + **cold water mass** has developed on the cavity side, sharply fronted against the 34–35 PSU open ocean, with the  $w$  runaway as a **vertical curtain at the  $-66^\circ$  front**. This is the clearest single view of the instability *amplifying in time* from a value-clean but dynamically-unbalanced start (§1). Frames from `blowup_section_evolution.mp4`.

### 3 Goal and setup

- **Experiment:** `movcav4`, an iterative `esm_tools` coupling. Model 1 = AWI-ESM3 (1-year chunks); Model 2 = Antarctic PISM (10-year chunks). The leg under test is **chunk-3**, AWI-ESM3 year 1901 — the *first* leg after a PISM step has changed the ice geometry.
- **Moving cavity:** PISM ice → FESOM sub-mesh (208 679 nodes vs. 211 567 on the full CORE3 mesh). A per-leg tool chain (`ice2fesom`, `ocp-tool`, `remap_restart`) regenerates the OASIS grids/masks/weights (A096 atmosphere grid = 40 320 points, `feom` ocean grid) and the OIFS ICMGG file (land-sea mask, soil type, geopotential, ...).
- **Paths:**  
`model /work/ab0246/a270092/model_codes/awiesm3-develop-is/oifs-48r1`  
`experiment /work/ab0246/a270092/pism_repro/experiments/movcav4.`

### 4 Coupling-glue performance: per-leg cost profiling and the `fesom2ice` speedup

A separate workstream from the two blockers, recorded here because it governs how fast the iterative campaign can iterate. On each leg the two *models* are cheap (~18 min for a FESOM coupled year, ~18 min for a 10-yr PISM chunk on 45 nodes), but the *coupling glue* between them dominated the per-leg wall clock. The worst offender was `couple_out_fesom` (the `fesom2ice` ocean → ice handoff): ~36 min on `movcav5`, run as a single `nproc:1` step while a 45-node allocation sat idle. Same method as the rest of this log — measure, then fix.

#### 4.1 Root cause: 564 redundant whole-file reads, and no interpolation at all

The handoff note blamed a FESOM-unstructured → lon/lat *interpolation*. The code refutes that: `couplings/utils/fesom_scalar_array_to_LonLat.py` does *no* horizontal regridding — its output is written on the native FESOM nodes (`mesh.x2/mesh.y2`). The cost was pure redundant I/O. A nested (`time × level`) loop called `pf.get_data(...)` ≈ 12 × 47 = 564 times *per variable*, and *every* call re-opened and re-read the entire ~262 MB yearly file and recomputed the full time-mean, just to keep one depth level. Worse, because `how="mean"` averages over all times *inside* each call, the outer time loop stored the *identical* annual-mean field into all 12 output slots.

#### 4.2 Fix 1 — read once, vectorise (output bit-identical)

Replaced the nested loop with a single `get_data(depth=None, how="mean")` read plus a vectorised per-level extraction (same `ind_for_depth` mapping, broadcast across time). Validated **bit-for-bit** against the old script (empty `cdo diffn`, identical byte size) and golden-equivalent. Standalone: `temp` 283 → ~10 s, `salt` 537 → ~13 s (~24–42×).

#### 4.3 Fix 2 — the whole pipeline ran twice

`iterative_coupling_fesom1x_ice_make_forcing` builds the entire forcing *twice* (a multiyear-mean pass and a total-mean pass); only the final CDO mean differs — the `*_goodgrid.nc` files the second pass rebuilds are byte-identical to the first. Guarded the `temp/salt` build loops to reuse the existing goodgrid when the raw source is unchanged (single-year legs only; rebuilds if the source is newer). *Fixes 1+2 committed:* `esm_tools` 53afb1f3, branch `development.interactiv_mesh.awiesm3`.

#### 4.4 Timer instrumentation ([TIMER])

To profile the glue in situ rather than by inference from file mtimes, a lightweight, idempotent lap-timer `couple_timer "label"` was added at the block boundaries of the five runtime-path coupling

files (31 marks total: `coupling_fesom2ice.functions`, `coupling_pism2esm.functions`, `coupling_pism2atmosphere.functions`, `coupling_ocean2pism.functions`, `coupling_atmosphere2pism.functions`). After any run,

```
grep '[TIMER]' <exp>/log/*couple*
```

prints a labelled per-step wall-clock breakdown (each mark reports the time since the previous one). Side-effect free, no behavioural change (74d67cc7).

#### 4.5 Ground truth from movcav6, and a corrected picture

The first instrumented leg (movcav6) confirms the win *and overturns the assumed bottleneck* (Table 1). Key readings: (i) **Fix 2 works** — `build_grid` pass 2 fell 140.9 → 2.6 s; (ii) `couple_out_fesom`

| Step (movcav6, real leg)                                      | Wall / s |
|---------------------------------------------------------------|----------|
| <i>couple_out_fesom (fesom2ice) — ~507 s total (~8.5 min)</i> |          |
| construct+select (cdo monmean) — pass 1                       | 158.96   |
| build_grid temp+salt — pass 1                                 | 140.86   |
| concat+means — pass 1                                         | 25.72    |
| construct+select (cdo monmean) — pass 2 ( <i>redundant</i> )  | 154.37   |
| build_grid temp+salt — pass 2 ( <i>now skipped, Fix 2</i> )   | 2.63     |
| concat+means — pass 2                                         | 23.60    |
| finalize + write grid/names                                   | 1.26     |
| <i>couple_in_pism — ~171 s total</i>                          |          |
| atmosphere2pism regrid+downscale                              | 19.83    |
| dEBM run (compile+execute)                                    | 64.81    |
| ocean2pism cdo remap (per-level weights, 47×)                 | 48.33    |
| ocean2pism extrapolate (fillmiss2)                            | 22.86    |
| (other atmosphere/ocean steps)                                | ~15      |
| <i>couple_out_pism (pism2esm) — ~119 s total</i>              |          |
| get_newest_output (cat+timmean, 120 steps)                    | 88.58    |
| ice-geometry sel timestep/expr                                | 24.14    |
| (orography/thickness anomalies, runoff, assemble)             | ~6       |

Table 1: movcav6 per-step coupling-glue timings from the [TIMER] instrumentation (`grep '[TIMER]'` on the `couple_*` logs).

overall dropped ~36 min (movcav5) → ~8.5 min; (iii) **the real #1 cost is `fesom.ice.select_relevant_variables` (the `cdo monmean`), ~155 s/pass — which the handoff note called “sub-second” — and it *also* runs twice (~310 s combined)**; (iv) `build_grid` in situ (141 s) is ~5.6× slower than the standalone ~25 s, i.e. live-filesystem contention, not the algorithm.

#### 4.6 Fix 3 — the `cdo monmean` was vestigial

The movcav6 table flagged `fesom.ice.select_relevant_variables` (a `cdo monmean` on each raw 3-D temp/salt file) as the #1 cost, ~155 s/pass × 2 passes ≈ 310 s. Tracing it shows the `*_var4ice.nc` it produces is **never read**: the Python builder reads the *raw* file directly and does its own time-mean, concatenate consumes the `*_goodgrid` files, and `var4ice` is used only to derive the `goodgrid filename` before being deleted. So the `monmean` — a full read+write of the ~260 MB yearly file — is pure overhead, left over from when the Python consumed `var4ice`. Replaced with an instant `ln -sf` (`goodgrid` name and `-f` invariants preserved; 2-D paths unchanged; 08ea580c). This removes both passes’ `monmean` and takes `couple_out_fesom` from ~508 s toward ~200 s — i.e. the `fesom2ice` handoff is now ~36 min → ~3–4 min across Fixes 1–3.

## 4.7 Remaining ranked targets

1. `pism2atmosphere get_newest_output (cat+timmean, ~89 s)` — largely *inherent*: 10 yr of monthly PISM extra output (120 steps,  $2.36 \times 10^9$  values). Low-yield/high-risk.
2. `ocean2pism cdo remap (~48 s)` — distance-weighted weights are regenerated *per depth level* ( $47\times$ ) because the wet/dry source mask varies by level; the source geometry is fixed, so the weights could be built once.
3. Minor: `pism2esm ice-geometry seltimestep (~24 s)`; `ocean2pism fillmiss2 (~23 s)`; a fixed `sleep 10` at the end of `ocean2pism`.
4. *Not* a target: `couple_in dEBM (~65 s)` — the “compile” is a fast `cp $DEBM_EXE`; the time is the dEBM Fortran model *executing* on a 1.1 GB input, i.e. genuine compute.

*Net:* the `fesom2ice` handoff is  $\sim 36$  min  $\rightarrow \sim 3$ –4 min (Fixes 1–3). The remaining PISM-side glue ( $\sim 8$  min, mostly inherent large-file CDO) is the next frontier if further speedup is wanted.

---

## Part II — The OIFS step-0 coastline crash (resolved; historical record)

---

*Everything from here to §13 documents the first blocker, now **solved**. It is kept as a full record of how the OIFS coastline crash was isolated and fixed; §2 supersedes it as the live state of the coupling.*

## 5 Infrastructure fixes already landed (coupling foundation)

Before the surface crash could be isolated, a series of framework/data bugs had to be cleared (all now resolved and validated at run time); these underpin *both* blockers:

- **OASIS weights/restarts:** fresh 208 679-point weights and remapped OASIS restarts are staged in the correct sequence; the MCT sparse-matrix dimension mismatch is gone. `rstos SST` is valid (271–305 K, no zeros); the `feom`  $\rightarrow$  A096 SST remap is exact (all 28 528 ocean cells row-sum = 1.0).
- **ICMGG regeneration:** the regenerated ICMGG land–sea mask equals the A096 OASIS mask *exactly* (11 792 land points, zero disagreement).
- **esm\_tools guards:** an `observe.py` race (compute launching before `couple_in` finished), `remap_oasis_restart` sanity guards, a shared `rmp-pool` write guard, and per-leg `plit` generation/wiring.

*Conclusion of this phase:* every component we *regenerate* agrees on the new coastline, so at that point the remaining crash was isolated to the OIFS side at initialisation — pursued through the rest of Part II.

## 6 Symptom: a moving-target step-0 crash

The leg dies during step 0, always on the same handful of MPI ranks — the ones holding the Antarctic coastline. As each cause was removed the crash moved *downstream*, which is itself the main diagnostic signal:

| Build          | Crash                       | Where / meaning                                                                                                                    |
|----------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| baseline       | error (73) divide-by-zero   | radaca: $\rho$ , PTH/PTH built from skin $T = 0$ .                                                                                 |
| +skin-T reseed | error (73) again            | skin $T$ re-zeroed: ocean scheme sets skin $T = \text{SST}$ , and SST was 0.                                                       |
| +SST reseed    | error (65) floating-invalid | advances past the whole surface/radiation chain; now NaN in cpdyddh:915 (DDH moist-physics budget, $\text{PQR} \times \Delta p$ ). |

So the surface divide-by-zero is *solved*; the live failure is a NaN precipitation flux produced by the moist physics on the same cells. (§8.5 later revised the *attribution* of the first two rows: the reseed acts only on NPROMA padding, and it is Approach A’s ICMGG re-ingest that actually populates the real flipped cells — but the observed crash-migration downstream is real either way.)

## 7 Method: measurement over speculation

Each of the findings below comes from an instrumented build (a debug WRITE at a specific line), run for five minutes on compute nodes, not from reading the code alone. The instrumentation lives behind the ECE LANDICE path so it does not touch ordinary runs.

## 8 What the measurements established

### 8.1 Skin temperature alone cannot be seeded

A step-0 reseed in updtim.F90 that set the skin-T reservoir to the freezing point RTT had *no effect*: the surface scheme re-derives ocean skin temperature from the SST later in the step, so with  $\text{SST} = 0$  the seed is overwritten back to zero before the radiation call. The SST field itself has to be seeded.

### 8.2 Seeding SST appears to clear the divide-by-zero

Adding an SST reseed ( $\text{SD\_VF\%YSST} \leftarrow \text{RTT}$  where  $< 100\text{K}$ ) coincided with the error (73) giving way to a downstream failure. Instrumentation showed no cell degenerate by the end of UPDTIM (DBG-UPDTIMEND: 0 cells still skinT(MP9)<100), and the model advanced into cpg\_drv → cpg → cpg\_dia before failing at

```
forrtl: error (65): floating invalid
cpdyddh_ 915 cpdyddh.F90 ! PDHCV(..)=PQR(JROF,JLEV)*ZDELPREV(..)
```

The invalid operand is a rain/snow flux PQR/PQS out of the moist physics — a genuine physics NaN, not a diagnostic artefact. (§8.5 shows the reseed was actually a no-op here; the real cells were already valid because of Approach A. The cpdyddh NaN is the true remaining failure regardless.)

### 8.3 The coupler delivers a valid SST to every ocean cell

The obvious next hypothesis was “the newly-wet cavity cells receive no SST from FESOM”. This was tested directly: in ece-updc1ie\_cp1.F90, at the exact line that applies the coupled SST to ocean-gated cells,

```
IF (SD_VF(JROF,YLSM%MP,IBL) <= 0.5 .AND. SD_VF(JROF,YCLK%MP,IBL) <= 0.5) THEN
  SD_VF(JROF,YSST%MP,IBL) = CPL_FLD_SST(IGP) ! <-- incoming FESOM SST
```

we counted, per rank, the ocean-gated cells whose *incoming* CPL\_FLD\_SST is degenerate ( $< 100$  K), printing each cell’s longitude, latitude, masks and `plit`. Result, run 26030286:

```
DBG-CPLSST-TOTAL: 0 ocean-gated cells got coupled SST<100 K, of NGPTOT=105    (x384 ranks
)
```

**Zero** on all 384 ranks. The coupler supplies a valid SST to every ocean cell. The “FESOM is not delivering SST” explanation is therefore **refuted**, and with it the idea that the fix lives on the ocean/OASIS side.

#### 8.4 The “degenerate cells” were NPROMA padding

A per-cell classifier (DBG-FLIP) in `updtim.F90` dumped, for every cell with a degenerate skin  $T$  or SST *before* any reseed, its `lon`, `lat`, `LSM`, `CLK`, `CI`, `SST`, skin  $T$  and `plit` — but guarded on the real-grid bound  $IGP \leq NGPTOT$ . The result was decisive:

```
DBG-FLIP-TOTAL: 0 real cells degenerate (skinT<100 OR SST<100) before reseed    (x384)
```

**Zero real cells.** The “ $\sim 7$  degenerate cells per rank” that the reseed counter had been reporting are the NPROMA *padding* slots beyond `NGPTOT` (which are legitimately zero); the reseed’s `DO JROF=1,NPROMA` loop counted them, the classifier’s `NGPTOT` guard excluded them. So the reseed is a no-op on the real grid, and the moist-physics NaN is *not* caused by a degenerate skin  $T$ /SST on any real cell. This reversed the working picture and forced the next question: what actually differs on the crash cells?

#### 8.5 The crash cells are exactly the cells re-ingested from ICMGG

The answer was found by realising the tree already contains an implementation of *Approach A* (§10): `reresf_part2.F90`, gated on `ECE_LANDICE`, reads the regenerated ICMGG with `SUGRIDG` on a warm start and re-applies its surface column — via a new masked `PUTALLFLDS_M` operator in `surface_fields_mix.F90` — **only at grid points whose land–sea type flipped** ( $|\text{LSM}_{\text{icmgg}} - \text{LSM}_{\text{srf}}| > 0.5$ ). Instrumenting it (DBG-REINGEST, run 26032081) shows it is *surgical and physically sane*:

```
DBG-REINGEST igp=41 lon=58.97 latsin=-0.919 LSMsrf=6.29E-2 LSMicm=1.00 sktICM=270.4
stl1ICM=270.3
DBG-REINGEST igp=63 lon= 0.00 latsin=-0.938 LSMsrf=1.00    LSMicm=0.37 sktICM=271.8
stl1ICM=271.8
... total 70 cells across 16 ranks:  49 sea->land,  21 land->sea
```

70 cells total — the actual coastline change — with injected skin/soil temperatures of 262–272 K (all physical, no zeros). And the ranks it touches (2155–2174) are *exactly* the ranks that then NaN in `cpdyddh`. This is why §8.3 and the padding finding both came back clean: the re-ingest had already given every real cell a valid surface column. The problem is not the surface *state* — it is that flipping a cell’s land–sea *class* at run time, most acutely the 49 sea → land cells where a warm moist marine atmospheric column is now over frozen land, is what the first moist-physics step turns into a NaN.

**Arbiter: the re-ingest is necessary, not harmful.** A build with the re-ingest *disabled* (`LLREINGEST_DBG=.FALSE.`) does *not* run further — it reverts to the *original error* (73) divide-by-zero in `radaca` (skin  $T = 0$ ). With the re-ingest enabled the model instead advances *past* the whole surface/radiation chain and dies one subsystem later in `cpdyddh`. So Approach A is doing real work: it repairs the surface so the flipped cells are no longer degenerate. What remained was a *second*, independent problem — pursued in §11 and §12.

## 9 Background: why the runtime mask can disagree with ICMGG

Independently of the crash mechanism, it is worth recording that on a warm start the OIFS runtime land–sea mask does *not* come from the regenerated ICMGG at all. The restart path (cnt3\_glo.F90) is:

```
IF (NCONF == 1 .OR. NCONF == 302) THEN
  CALL RERESF_PART1 (...) ! sets LNF from restart-control file: present -> LNF=.FALSE
  .
  CALL RERESF_PART2 (...) ! srf bulk restore (GPOPER 'PUTALLFLDS'): ALL slots incl.
    LSM
ELSE
  LNF = .TRUE.
ENDIF
IF (LNF) CALL CSTA(...) ! ICMGG read (SUINIF/SUGRIDF/SUGRIDG); ONLY on a cold start
```

PUTALLFLDS (surface\_fields\_mix.F90) copies every NDIM storage slot with no per-field flag, so the land–sea mask is always overwritten from `srf`; and because a `srf` restart is present each leg, `LNF=.FALSE`. and `CSTA` — the only reader of the regenerated ICMGG geometry — never runs. A KTRAJ tagging attempt in `su_surf_flds.F90` was ruled out: KTRAJ controls the trajectory/FullPos write, not the bulk restore, so it had no effect on the runtime mask. This stock behaviour — a stale, `srf`-restored mask on every warm start — is precisely what Approach A (§10, now implemented in `RERESF_PART2`) was written to override at the flipped cells.

## 10 The two approaches, and where they now stand

Two designs can make OIFS respect the moving coastline. §8.5 showed the first is *already built and running*; the open question is no longer “which to implement” but “is the implemented one physically viable, or must we fall back”.

**Approach A — re-ingest the regenerated ICMGG geometry on a warm start** (*implemented, live*). `reressf_part2.F90` (gated on `ECE_LANDICE`) reads the regenerated ICMGG via `SUGRIDG` and, using the new masked `PUTALLFLDS_M` operator (`surface_fields_mix.F90`), replaces the *entire* surface column at the 70 cells whose land–sea type flipped, leaving every other cell on its spun-up `srf` state. §8.5 confirmed it does exactly this, with sane values. *The concern it raises*: it changes a cell’s land–sea class between two model states in a single step. The atmospheric column above a sea → land cell is still a marine column; the surface beneath it is now frozen land. The first moist-physics step over that mismatch is the current NaN suspect. If the disabled-re-ingest run comes back clean, this is confirmed and Approach A needs either a gentler transition (e.g. only flip where `plit` also marks ice, so the cell enters the *glacier* path rather than bare-soil land) or a matching adjustment of the near-surface atmosphere.

**Approach B — the established `plit` overlay.** EC-Earth’s ice-sheet coupling (documented in `ece4-pism-coupling.md`) does *not* re-read or flip the LSM. It keeps the `srf` geometry and communicates the evolving ice as a fractional mask `plit`: `suorog.F90` patches the orography from `usurf_oifs` and (with `ECE_LANDICE`) reads `plit_oifs` at leg init, and the live `PLIT<region>` OASIS field drives the *glacier* physics via `SURFACE.GET_LANDICE`. *Pros*: it is the design the OIFS ice code was written for, and it never creates the abrupt land–sea-class flip that Approach A does. *Cons*: it depends on `plit_oifs` being generated (`make_plit_oifs.py`), read and applied, and on the live `PLIT` path (the ISM-mapper OASIS component), which the `movcav4` setup may not yet run.

**Where the evidence points.** Approach A is confirmed necessary (§8.5), and it is kept. But it faithfully transfers whatever the coupling tool chain writes into the ICMGG — so if a flipped cell is

inconsistent there, the re-ingest dutifully installs that inconsistency at run time. That turned out to be the actual bug (§11), and it lives in the tool world, not in OIFS.

## 11 The real root cause: `ocp-tool` leaves flipped cells half-converted

An FPE-safe operand check at the crash line (DBG-DDH) showed the non-finite operand is PCOVPTOT (the precipitation-cover fraction from the moist physics), not the pressure thickness. Following that upstream to the surface state of the crashing cell (a sea → land cell at 74.3°S, 257°E) exposed the flaw — in the regenerated ICMGG it reads

```
lsm=1.00  slt=6  swvl1=0.000  stl1=272.0  ci=0.838  sst=270.6  <-- half ocean, half
land
```

i.e. a *land* mask over an otherwise *ocean* column: zero soil moisture, a leftover sea-ice fraction and an SST. `ocp-tool`'s `lsm.py` (`modify_lsm`) flips a cell by setting *only* the land-sea mask and the soil type (`slt=6` = “sandy clay loam”), and leaves *every other surface field* at its old-type value. The moist physics over that mongrel column manufactures the NaN.

**Fix (tool world, committable).** `modify_lsm` now rebuilds the *whole* surface column of each flipped cell from the nearest stable neighbour of its *new* type (a masked great-circle nearest-neighbour fill, `fill_flipped_from_nearest_neighbour`). A newly-grounded cell then looks like the ice/land around it (`slt=1`, `swvl1`≈0.30, `ci`= 0, glacier snow), a newly-wet cell like the ocean around it — with the freshly set land-sea mask preserved. Regenerating this leg's ICMGG with the fix turns the crash cell from the mongrel column above into `lsm=1`, `slt=1`, `swvl1=0.302`, `ci=0.000` — a self-consistent glacier-land column. This is the right home for the fix: OIFS stays generic and the coupling machinery owns coastline consistency.

## 12 The residual `cpdyddh` NaN was DDH-diagnostic-only

With the consistent ICMGG, the model still stopped at `cpdyddh` — but that routine is a *DDH budget diagnostic* (Diagnostics of Horizontal Domains), and the crash line accumulates a *passive* precip-cover variable. Disabling DDH in the namelist (`NAMDDH: LHDGLB=LHDPRG=LHDHKS=.false.`) removed it, and OIFS then ran **~700 coupling steps** without any FPE trap. Since OIFS is compiled `-fpe0` (every `arpifs` object carries `-fpe0 -fp-model precise`), running that far proves its forecast state — and the fluxes it sends the ocean — are finite: the PCOVPTOT NaN was confined to the DDH passive-variable accumulation (an uninitialised diagnostic slot at step 0), not the real physics. OIFS is no longer the blocker.



## 13 OIFS crash — final status (resolved)

| Item                                           | Status                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coupling machinery (mesh, OA-SIS, ICMGG regen) | <b>Consistent &amp; validated</b> ; ICMGG mask = A096 mask exactly; ocean SST delivery 0/384 degenerate.                                                                                                                                                                                                                                                                   |
| OIFS surface divide-by-zero                    | <b>Solved</b> by the ICMGG re-ingest (Approach A, RERESF_PART2), confirmed necessary by the disable arbiter.                                                                                                                                                                                                                                                               |
| Flipped-cell inconsistency (ocp-tool)          | <b>Fixed</b> — masked nearest-neighbour rebuild of each flipped cell’s surface column (§11); verified in the regenerated ICMGG.                                                                                                                                                                                                                                            |
| cpdyddh NaN                                    | <b>Diagnostic-only</b> ; disabling DDH (NAMDDH) let OIFS run ~700 coupling steps under -fpe0. Real physics/fluxes are finite (§12).                                                                                                                                                                                                                                        |
| <b>OIFS overall</b>                            | <b>No longer the blocker.</b>                                                                                                                                                                                                                                                                                                                                              |
| <b>New frontier: FESOM</b>                     | <b>Superseded by §2</b> (the current truth). This row’s open question — “finite-but-extreme OIFS forcing” — has since been <b>ruled out</b> (§2): the blowup is a <i>time-growing coastal baroclinic instability</i> on the submesh, ocean-internal, driven by subsurface freshening at the retreated margin; timestep, flux, ice and restart corruption are all excluded. |

*Key files:* `ocp-tool/lsm.py` (`modify_lsm + fill_flipped_from_nearest_neighbour` — the tool-world fix), `arpifs/control/eresf_part2.F90` + `module/surface_fields_mix.F90` (re-ingest PUTALLFLDS\_M), `arpifs/dia/cpdyddh.F90` (DDH crash site), `NAMDDH` in `fort.4` (DDH off). Debug instrumentation (DBG-\*) in `updtim.F90`, `ece_updclie_cpl.F90`, `climaer_layer.F90`, `cpdyddh.F90` is to be stripped before committing OIFS.

## 14 To do

- ☐ **ocp-tool NN fix** — *done*: committed + pushed, PR #51 (JanStreffing/ocp-tool, branch `fix/flipped-cell-nn-fill` → master).
- ☐ **Commit the OIFS re-ingest** (branch `movcav-landice`, SMHI fork). First clean it up:
  - *keep*: `surface_fields_mix.F90` (PUTALLFLDS\_M, no debug), `bundle.yml`;
  - *strip*: the DBG-REINGEST block and the LLREINGEST\_DBG toggle in `eresf_part2.F90` — make the re-ingest unconditional under `ECE.LANDICE`;
  - *revert entirely* (pure debug / no-op): `updtim.F90` (reseed + DBG-FLIP), `ece_updclie_cpl.F90` (DBG-CPLSST), `climaer_layer.F90` (DBG-CLIMAER), `cpdyddh.F90` (DBG-DDH).
- ☐ **Make DDH-off permanent** in `esm_tools` — fold the `NAMDDH LHDGLB=LHDPRG=LHDHKS=.false.` into the OIFS namelist config (likely `configs/components/oifs/oifs48.cmip.yaml`) rather than the runtime `fort.4`.
- ☐ **Curate + commit the esm\_tools WIP** (branch `development_interactiv_mesh_awiesm3`): the OASIS-regen/plit wiring in `coupling_ice2fesom_interactive_mesh.functions`, the `remap_oasis_restart.py` guard, the `observe.py` race fix, the `awiesm3.yaml` pool guard — split into logical commits.
- ☐ **Root-cause the live FESOM instability** (§2). The frontier is no longer OIFS: it is a time-growing coastal baroclinic mode fed by margin freshening. Next probes, in order: (a) the sub-shelf **meltwater/freshwater flux** on the submesh where the cavity was just exposed; (b) submesh **element quality** / `zbar-hnode consistency` at the new coastline; (c) whether

the freshening is surface meltwater mixed down or a distributed sub-shelf source (section movie done, §2.3).